ADITYA COLLEGE OF ENGINEERING AND TECHNOLOGY
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Affiliated to VTU, Belagavi | Approved by UGC/AICTE/Govt. of Karnataka, Bengaluru

DEPARTMENT OF PHYSICS
APPLIED PHYSICS FOR CSE STREAM
QUESTION BANK

Sl. NO	MODULE-I LASER AND OPTICAL FIBERS
1	Write a short note on • Spontaneous and stimulated emission • Requisites and conditions of Laser • How matter interacts with radiation
2	Define the following terms • Population Inversion • Meta stable state • Acceptance angle • Numerical aperture • Fractional index change
3	Derive an expression for numerical aperture of an optical fibre with a suitable diagram
4	Explain the various types of optical fibres with proper diagrams, also describe how P-P communication is enabled through optical fibres
5	What is meant by attenuation and attenuation coefficient, Mention the expression for attenuation coefficient in a fibre of length L, Mention advantages and disadvantage
6	Derive the expression for energy density of radiation using Einstein's coefficients, Compare the expression with Planck's Equation
	MODLUE-II QUNAMTUM MECHANICS
1	Setup time independent Schrodinger's wave equation
2	Explain de-Broglie Hypothesis, derive an expression for de-Broglie Wavelength
3	State and explain the Heisenberg Uncertainty principle with relevant equations, based on the principle, prove that free electrons do not exist inside the nucleus.
4	What is wave function? Give its physical significance and properties
5	Discuss the wave functions, probability densities and energy level for a particle in a box by considering the ground and the first two excited State
6	What are the properties of a wave function? Give the qualitative explanation of Max-Born's interpretation of wave function
7	What are matter waves, mention their properties, Define phase and Group Velocity
8	Obtain the expression for Eigenvalue and Eigenfunction for particles in a box.
	MODLUE-III QUANTUM COMPUTING
1	Explain the representation of qubit using Bloch Sphere.
2	Explain Moore's law & its end.
3	Describe the working of controlled% gate mentioning its matrix representation and truth table.
4	Elucidate the differences between classical and quantum computing
5	State the Pauli matrices and apply Pauli matrices on the states |0> and |l>.
6	Discuss the CNOT gate and its operation on four different input states
7	Explain the Matrix representation of 0 and 1 States and apply identity operator I to |0> and |l> states
8	Define a qubit and explain the properties of qubit
9	Explain Pauli X, Y and Z gate on the states 0 and 1 with proper gate representations and their truth table.
10	Discuss the working of phase gate mentioning its matrix representation and truth table.

11	Explain Orthogonality and Orthonormality with an example for each.
12	Explain Single qubit gate and multiple qubit gate with an example for each.
	MODULE-IV **ELECTRICAL PROPERTIES OF MATERIALS AND APPLICATIONS**
1	Write the assumptions of quantum free electron theory. Discuss about the success (merits) of quantum free electron theory.
2	Describe how quantum free electron theory has been successful in overcoming the failures of classical free electron theory.
3	Discuss the merits of quantum free electron theory.
4	Explain Fermi energy and Fermi factor. Discuss the variation of Fermi energy with temperature
5	Explain classical free electron theory of metals. Discuss in details about the failures of classical free electron theory,
6	Discuss about the concept of phonon and Matthiessen's rule.
7	Explain in detail about Type-I and Type-Il superconductors.
8	Explain Meissner's Effect and the variation of critical field with temperature
9	Explain the phenomenon of superconductivity and discuss qualitatively the BCS theory of superconductivity for negligible resistance of metal at temperatures close to absolute zero.
10	Explain DC and AC Josephson effects and mention the applications of superconductivity in quantum computing.
11	Describe Meissner's Effect and hence classify superconductors into Soft and Hard superconductors using M-H graphs.
	MODULE-V **PHYSICS OF ANIMATION**
1	Discuss timing in linear motion, Uniform motion, slow in and slow out
2	Sketch and explain the motion graphs for linear, easy ease, easy ease in and easy ease out cases of animation.
3	Distinguish between descriptive and inferential statistics.
4	Illustrate the odd rule and odd rule multipliers with a suitable example
5	Describe Jumping and parts of jump
6	Discuss the salient features of Normal distribution using bell curves,
7	Elucidate the importance of size & scale and weight and strength in animations.
8	Mention the general pattern of Montecarlo method and hence determine the value of π
9	Describe the calculation of Push time and stop time with examples.
10	Discuss Poisson distribution and modelling the probability for proton decay.
11	Define a frame and show the importance of Frames per second in animation.

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ALL THE BEST

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